

Daniel English-Brown

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SUMMARY

MSc Theoretical Physics student at King's College London with growing specialisation in quantum information theory, quantum error correction and quantum algorithms. Experienced in Python-based numerical modelling, stabiliser-code analysis, with current projects on topological quantum codes and the Quantum Approximate Optimisation Algorithm (QAOA). Seeking PhD opportunities in theoretical quantum computing/quantum information.

RESEARCH AND PROFESSIONAL EXPERIENCE

Lead Web Developer, Expendid

Sept 2025–Present

- Designed and implemented the full-stack architecture of *Expendid*, an AI-driven personal finance application.
- Developed a modular Python–Flask backend and responsive JavaScript frontend supporting automated data pipelines and user analytics.
- Applied numerical-optimisation routines to refine expenditure-prediction algorithms used in the product's beta version.

Quantitative Research Intern, Perbak Capital Partners (London)

Sept 2024

- Constructed and validated quantitative pricing models (Black–Scholes, binomial tree) for derivative valuation and volatility estimation.
- Engaged in structured study of CVPXY and asset-pricing theory (CAPM, volatility modelling).
- Worked with quantitative researchers through risk-neutral pricing theory.

STEM Outreach Ambassador, King's College London

Sept 2023–Present

- Delivered public demonstrations of physical phenomena, fostering engagement with electromagnetism, optics, and quantum mechanics.
- Presented the NMES faculty to diverse audiences, promoting accessibility of scientific research and higher education.

SELECTED RESEARCH PROJECTS

Implementing and Analyzing the Quantum Approximate Optimization Algorithm (QAOA) *In Progress — Expected Jan 2026*

Supervised by Prof. Nikolay Gromov. Developing a full QAOA implementation in Qiskit to study Max-Cut and related NP-hard problems. Constructing parameterised quantum circuits, analysing optimiser performance, mapping parameter landscapes, and benchmarking algorithm robustness using realistic IBM

noise models and error-mitigation techniques.

Comparative Study of Topological Codes: Surface, Colour, and Hyperbolic Variants *In Progress — Expected Sep 2026*

Supervised by Dr. Po-Shen Hsin. Theoretical comparison of code rate, distance, and stabiliser structure across multiple topological quantum error-correcting codes. Studying how lattice geometry influences thresholds and logical-gate construction, with emphasis on hyperbolic codes and connections to higher-dimensional topological field theories.

Modelling a Ring Resonator Using a 2-Dimensional Finite Difference Time Domain (FDTD) Software *Completed May 2025 — First-Class Grade*

Supervised by Dr. F. Rodríguez-Fortuño. Built a 2D FDTD solver in Python to model dielectric ring resonators using the Yee lattice and tapered absorbing boundaries. Analysed spectral modes, free spectral range, and quality factors, validating theoretical predictions.

EDUCATION

- 2025–2026 **MSc Theoretical Physics**, King’s College London (*Expected Distinction*)
- 2022–2025 **BSc (Hons) Theoretical Physics**, King’s College London
- 2020–2022 **A-Levels**, Ampleforth College — Physics, Chemistry, Mathematics, French (AABB); Academic Scholarship.

TECHNICAL AND ANALYTICAL SKILLS

Programming	Python (NumPy, SciPy, Pandas, Matplotlib, CVXPY), Qiskit, Mathematica, MATLAB, JavaScript, HTML/CSS, LaTeX, Git
Numerical Methods	Finite-Difference Methods (FDTD, ODE/PDE solvers), Optimisation Algorithms, Eigenvalue Problems, Spectral Analysis
Data & Analysis	Statistical Modelling, Regression, Signal Processing, Data Visualisation
Languages	English (C2), Mandarin (C1), French (B2)
Soft Skills	Technical Communication, Research Presentation, Scientific Writing

ACHIEVEMENTS AND TEST RESULTS

SAT: 1510/1600 (99th percentile) | **HSK 5 (Mandarin):** 179/300 | **Academic Scholarship:** Ampleforth College